



Toni Blair

Software Engineer | Full-Stack Developer | Product & Content System
4th year Dual Degree Student | IIT Kharagpur

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EDUCATION

2023 - present	Indian Institute of Technology, Kharagpur	7.4	Dual Degree: Ocean Engg & Naval Arch 12th
2023	Tata DAV school, Jamadoba(DHN)	83.30%	CBSE Board
2021	Denobili School F.R.I, Dighwadih	85.00%	10th ICSE Board

SKILLS AND EXPERTISE

Languages: Python, JavaScript, TypeScript, C++, SQL | **Product & Design:** Product Design, UI/UX, Interaction Design, Information Architecture, User Flows, Prototyping, Figma, Design Systems, Design-to-Code | **Frontend:** React.js, React Native, Next.js, HTML5, CSS3, Tailwind CSS, Expo | **Backend:** FastAPI, Node.js, Express.js, REST APIs, WebSockets, JWT Authentication | **AI & APIs:** Generative AI, Google Gemini, AI-Powered UX, Agentic Development, Prompt Engineering, LLM/API Integration, Third-Party API Integration | **Databases:** PostgreSQL, MongoDB, SQLite, Redis | **Tools:** Git, GitHub, Docker, Linux, VS Code, Postman, Vercel, Cloudinary | **Concepts:** Full-Stack Development, Component Architecture, API Design, Data Structures & Algorithms, OOP, DBMS

PROJECTS

Chalo: Hyperlocal Activity Platform | *React Native, FastAPI, PostgreSQL, Async SQLAlchemy, JWT, Docker*

- Building a mobile platform for discovering and joining nearby activities, with a React Native frontend and FastAPI backend organized around separate application modules.
- Implemented backend foundations including JWT-based authentication, asynchronous database workflows with PostgreSQL/SQLAlchemy, OTP generation and expiry handling, and API endpoints for user/activity workflows.
- Designed the project with a modular client-server architecture and Dockerized backend setup; actively extending features including verification, onboarding and activity discovery.

Wyrm Store: Full-Stack E-Commerce Platform | *React (Vite), Node.js, Express.js, MongoDB, JWT, Cloudinary*

- Built an end-to-end e-commerce application with secure authentication, role-based admin controls, product catalog, cart, checkout and order management workflows.
- Developed REST APIs for authentication, products, carts, orders and administrative operations, including user/role management, order status updates and sales analytics.
- Implemented image upload support through Cloudinary and built protected frontend routes for customer and

Multithreaded HTTP Server | *C, POSIX Sockets, pthreads, HTTP, Linux, Make*

- Built a lightweight HTTP server from scratch using TCP sockets and POSIX threads, accepting client connections and handling requests concurrently with a thread-per-client model.
- Implemented manual HTTP request parsing and response construction, static file serving, Content-Type handling and 404 responses.
- Created a Makefile-based build workflow and documented architecture and current limitations, including the next-step improvement of introducing a bounded thread pool.

TB Scout: Venture Intelligence Platform | *Next.js, TypeScript, Zustand, Tailwind CSS* | *Product Design & Development*

- Designed a research workflow for discovering and evaluating startups across country, sector, stage, valuation, funding, and growth signals turning a large dataset into a focused decision-making interface.
- Built interactive company profiles, valuation-growth visualizations, saved research, diligence notes, target lists, filtering, sorting, and CSV export to support an end-to-end research workflow.
- Implemented live domain enrichment and persistent client-side state, with reusable interface patterns for complex information-dense product surfaces.

POSITIONS OF RESPONSIBILITY

Design Sub-Head | Samudramanthan (OENA Dept Fest), IIT Kharagpur

November '24 - present

- Led the design vertical for the OENA departmental fest; created posters, banners, ID cards, certificates and digital creatives.

Maintainance & Socult Secretary | Rajendra Prasad Hall, IIT Kharagpur

August '24 – May '25

- Coordinated hall maintenance and student welfare; contributed to hall merchandise and design initiatives.

LEADERSHIP & ACHIEVEMENT

- Gold Medal (Team) --- Sketching Event, Inter-Hall General Championship, IIT Kharagpur.
- Design Team Member --- Finess (Fashion Design), Inter IIT Cultural Meet 7.0.
- NSO Coordinator --- Hockey & Health and Fitness, IIT Kharagpur.